Ex 9.1.1

Prove Proposition 9.2. (Hint: Either use Proposition 9.3 or use the observation that the formula (9.4) holds for all circles γ_τ with τ sufficiently small.)

Proposition 9.2. Suppose that f has an isolated singularity at a with associated Laurent series

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{j=1}^{\infty} \frac{b_j}{(z-a)^j}.$$

(a) a is a removable singularity if and only if all the b_j are zero.

(b) a is a pole of order $m \geqslant 1$ if and only if $b_m \neq 0$ and $b_j = 0$ for $j > m$.

(c) a is an essential singularity if and only if $b_j \neq 0$ for infinitely many j.

Pf. We have that the Laurent series above is valid on a punctured disk, $0 < |z-a| < r$. We use Thm 9.1's (Laurent's Theorem) uniqueness statement to say that the coefficients a_n, b_j are determined by f.

(a) ($\Leftarrow$) Let $\forall b_j = 0$, then $f(z) = \sum_{n \geqslant 0} a_n (z-a)^n$ on the punctured disk. Using Thm 1.34, this power series defines a holomorphic function on the full disk $D_r(a)$, therefore setting $f(a) = a_0$ makes f holomorphic at a. Therefore, by Def 5.12, the singularity is removable.

($\Rightarrow$) If a is removable, let g be the holomorphic extension of f to $D_r(a)$. By Cor 4.14 (Taylor's Theorem), we have on $D_r(a)$,

$$g(z) = \sum_{n \geqslant 0} c_n (z-a)^n$$

which is the Laurent series for f on $0 < |z-a| < r$ whose negative coefficients all vanish. Therefore, using uniquness in Thm 9.1 we must have $\forall b_j = 0$ and $a_n = c_n$. □

(b) ($\Leftarrow$) Let $b_m \neq 0$ and $b_j = 0$ for $j > m$. Then, we have,

$$f(z) - \sum_{j=1}^{m} \frac{b_j}{(z-a)^j} = \sum_{n \geqslant 0} a_n (z-a)^n$$

which is known to be removable by part (a). By Def 5.14, we have exactly that the a is pole of order m.

($\Rightarrow$) Let a be a pole of order m. By Def 5.14, we have $\exists c_1, \ldots, c_m$ with $c_m \neq 0$ s.t.

$$r(z) = f(z) - \sum_{j=1}^{m} c_j (z-a)^{-j}$$

has a removable singularity at a. Using part (a), the Laurent series of r has no principal part, therefore,

$$f(z) = \sum_{n \geqslant 0} a_n (z_a)^n + \sum_{j=1}^{m} \frac{c_j}{(z-a)^j}$$

By Thm 9.1's uniqueness, $\forall j, 1 \leqslant j \leqslant m : b_j = c_j$ and $\forall j > m : b_j = 0$. I.e., we have $b_m = c_m \neq 0$. □

(c) ($\Rightarrow$) By Def 5.20, a is essential when it is neither removable nor a pole. By part (a), if a is removable then $\forall b_j = 0$, and by part (b) if a is a pole then it has the largest index m with $b_m \neq 0$ s.t. $\forall j > m : b_j = 0$. In either case, only finitely many b_j are nonzero.

($\Leftarrow$) Ff only finitely many b_j aare nonzero, then either all vanish (removabel) or there is a largest m with $b_m \neq 0$ (pole of order m).

Therefore, a is essential iff infinitely many b_j are nonzero. □

Ex 9.2.2

Suppose that f has an isolated singularity at a. Show that if $\lim_{z\to a}(z-a)f(z) = L \neq 0$, then f has a simple pole at a with residue L. What happens if $L = 0$?

Pf. On the punctured disk where f, we consider $\varphi(z) = (z-a)f(z)$. We can see that $\lim_{z\to a}\varphi(z) = L$ exists and is finite (since φ is bounded near a). This fulfills Prop 5.13's boundless criterion (d), since the limit exists, which gives us (a) i.e. the singularity of φ at a is removable.

We can then extend φ holomorphically s.t. $\varphi(a) = L$, and can write its Taylor series as,

$$\varphi(z) = \sum_{n\geqslant 0} c_n(z-a)^n$$

Therefore, $c_0 = \varphi(a) = L$. On the punctured disk, we have,

$$f(z) = \frac{\varphi(z)}{z-a} = \frac{L}{z-a} + \sum_{n\geqslant 1} c_n(z-a)^{n-1}$$

where the second term is a power series and holomorphic at a. Therefore, the Laurent series of f at a has $b_1 = L \neq 0$ and $\forall j \geqslant 2 : b_j = 0$.

Using Prop 9.2 (b), proven above in Ex 9.1.1, we have a is a pole of order 1, i.e. a simple pole. By Def 9.6 we also have that the residue is $\operatorname{Res}(f; a) = b_1 = L$. □

In the case $L = 0$: using the same procedure gives φ holomorphic at a with $\varphi(a) = 0$. Therefore, we have $\varphi(z) = \sum_{n\geqslant 1} c_n(z-a)^n$ and that,

$$f(z) = \frac{\varphi(z)}{z-a} = \sum_{n\geqslant 1} c_n(z-a)^{n-1} = \sum_{m\geqslant 0} c_{m+1}(z-a)^m$$

is a power series. Therefore f has a removable singularity at a, i.e. a is not a pole. So when $L = 0$ the only possibility is that the singularity is removable.

Ex 9.2.3

Prove the Simple Pole Lemma on page 138.

Lemma 9.16 (Simple Pole Lemma). Suppose that g and h are holomorphic at a, that $g(a) \neq 0$ and that h has a simple zero at a. Then $f = g/h$ has a simple pole at a and

$$\operatorname{Res}(f; a) = \frac{g(a)}{h'(a)}.$$

Pf. Since h has a simple zero (a zero of order 1) at a, we have from Lemm 5.4 that the function h_1 is holomorphic at a with $h_1(a) \neq 0$ and that,

$$h(z) = (z - a)\, h_1(z).$$

We can differentiate to get,

$$h'(z) = h_1(z) + (z - a)h_1'(z)$$

Therefore, $h'(a) = h_1(a) \neq 0$. Since $h_1(a) \neq 0$ and $g(a) \neq 0$, we have a punctured disk on which h_1, and hence h, are nonzero. Therefore, $f = g/h$ is holomorphic there and has an isolated singularity at a. Then,

$$\lim_{z\to a}(z - a)f(z) = \lim_{z\to a} \frac{(z - a)g(z)}{(z - a)h_1(z)} = \lim_{z\to a} \frac{g(z)}{h_1(z)} = \frac{g(a)}{h_1(a)} = \frac{g(a)}{h'(a)} \neq 0$$

From Ex 9.2.2 above we have that f has a simple pole at a with residue equal to this limit, i.e. $\operatorname{Res}(f; a) = \dfrac{g(a)}{h'(a)}$. $\square$

Ex 9.2.5

Suppose that f has a pole of order m at z_0 and that k has a zero of order n at z_0. Show that $\operatorname{Res}(f'/f; z_0) = -m$ and that $\operatorname{Res}(k'/k; z_0) = n$. (Compare with Ex 5.2.7.)

Pf. Since k has a zero of order n at z_0, using Lemm 5.4 gives a function K holomorphic at z_0 with $K(z_0) \neq 0$ and $k(z) = (z - z_0)^n K(z)$. Then we can differentiate to get,

$$k'(z) = n(z - z_0)^{n-1} K(z) + (z - z_0)^n K'(z)$$

which gives,

$$\frac{k'(z)}{k(z)} = \frac{n}{z - z_0} + \frac{K'(z)}{K(z)}$$

Since $K(z_0) \neq 0$, the term K'/K is holomorphic near z_0. Hence k'/k has a simple pole at z_0 (as in Ex 5.2.7). We can use $\lim_{z \to z_0} (z - z_0)\, k'/k = n \neq 0$ with Ex 9.2.2 to obtain,

$$\operatorname{Res}\left(\frac{k'}{k}; z_0\right) = n.$$

Since f has a pole of order m at z_0, Cor 5.18 gives a function F holomorphic at z_0 with $F(z_0) \neq 0$ and $f(z) = F(z)/(z - z_0)^m$. Therefore we have,

$$\begin{aligned} f'(z) &= \frac{F'(z)}{(z - z_0)^m} - \frac{mF(z)}{(z - z_0)^{m+1}} \\ \frac{f'(z)}{f(z)} &= \frac{F'(z)}{F(z)} - \frac{m}{z - z_0}. \end{aligned}$$

Since $F(z_0) \neq 0$, F'/F is holomorphic near z_0. Hence f'/f has a simple pole at z_0, and its residue is the coefficient of $(z - z_0)^{-1}$, i.e. $\operatorname{Res}\left(\frac{f'}{f}; z_0\right) = -m$. □

Ex 9.3.5

Show that
$$I = \int_0^{2\pi} \cos(\theta)^{2n}\, d\theta = \frac{\pi(2n)!}{2^{2n-1}(n!)^2}.$$

Pf. Like in Section 9.3.1, parametrize $z = e^{i\theta}$, s.t. $\theta \mapsto z$ traverses the positively oriented unit circle γ, with $dz = iz\, d\theta$ (hence $d\theta = dz/(iz)$) and $\cos\theta = \frac{1}{2}(z + z^{-1})$. Therefore we have,

$$I = \oint_\gamma \left(\frac{z+z^{-1}}{2}\right)^{2n} \frac{dz}{iz} = \frac{1}{2^{2n}i} \oint_\gamma \frac{(z+z^{-1})^{2n}}{z}\, dz$$

Using the binomial theorem, we have,

$$(z+z^{-1})^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} z^{2k-2n}$$

therefore,

$$\frac{(z+z^{-1})^{2n}}{z} = \sum_{k=0}^{2n} \binom{2n}{k} z^{2k-2n-1}$$

We have that the only singularity of the integrand is at 0, and its residue is the coefficient of z^{-1}, given by $2k - 2n - 1 = -1 \Rightarrow k = n$:
$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}$$

We have that γ is a positively-oriented Jordan path and the integrand is holomorphic inside (besides at 0), we have using Cor 9.13 (the Residue Theorem) that $\oint_\gamma = 2\pi i \binom{2n}{n}$. Therefore,

$$I = \frac{1}{2^{2n}i} 2\pi i \frac{(2n)!}{(n!)^2} = \frac{2\pi}{2^{2n}} \frac{(2n)!}{(n!)^2} = \frac{\pi(2n)!}{2^{2n-1}(n!)^2}$$

as required. □

Ex 10.1.1

Suppose that f is holomorphic on the punctured complex plane $D^0 = \mathbb{C} \setminus \{0\}$. Suppose there is a sequence of polynomials (p_n) converging to f uniformly on compact subsets of D^0. Show that f has a removable singularity at 0.

Pf. In $D^0 = \mathbb{C} \setminus \{0\}$, deleted neighborhood of 0, we have that 0 is an isolated singularity of f. Using Thm 9.1 (Laurent's Theorem), for $0 < |z| < \infty$, f has a Laurent series,

$$f(z) = \sum_{n \geqslant 0} a_n z^n + \sum_{j \geqslant 1} \frac{b_j}{z^j}$$

We can find its coefficients using equation (9.4), $\forall \rho > 0$ and $\forall j \geqslant 1$,

$$b_j = \frac{1}{2\pi i} \oint_{\gamma_\rho} f(w)\, w^{j-1}\, dw$$

where γ_ρ is the positively-oriented circle $|w| = \rho$.

We can consider $\rho > 0$, s.t. the circle γ_ρ^* is a compact subset of D^0. Therefore, $p_n \to f$ uniformly there. Since w^{j-1} is bound on γ_ρ^* we have $p_n(w)w^{j-1} \to f(w)w^{j-1}$ uniformly on γ_ρ^*.

Using Prop 3.26(b) we can pass to the limit inside the integral s.t.

$$b_j = \lim_{n \to \infty} \frac{1}{2\pi i} \oint_{\gamma_\rho} p_n(w)\, w^{j-1}\, dw$$

We can consider $j \geqslant 1$, s.t. the integrand $p_n(w)w^{j-1}$ is a polynomial. Using Lemm 4.5 (polynomial has an antiderivative, and γ_ρ is closed), its integral over γ_ρ is 0. Therefore, $\forall j \geqslant 1 : b_j = 0$.

Using Prop 9.2(a) (proved in Ex 9.1.1 above), the singularity of f at 0 is removable. □

Ex 9.3.10

Evaluate
$$\int_0^{2\pi} \frac{\cos(\theta)}{3+\cos(\theta)}\, d\theta.$$

Pf. We can rewrite the integral as $\dfrac{\cos\theta}{3+\cos\theta} = 1 - \dfrac{3}{3+\cos\theta}$, s.t.

$$J := \int_0^{2\pi} \frac{d\theta}{3+\cos\theta}$$

$$I = \int_0^{2\pi} \frac{\cos\theta}{3+\cos\theta}\, d\theta = \int_0^{2\pi} d\theta - 3\int_0^{2\pi} \frac{d\theta}{3+\cos\theta} = 2\pi - 3J$$

For J we parametrize $z = e^{i\theta}$, s.t. $d\theta = dz/(iz)$, $\cos\theta = \frac{1}{2}(z+z^{-1})$ and $3+\cos\theta = \dfrac{z^2+6z+1}{2z}$. Then we can consider it over the positively oriented unit circle γ so that we have,

$$J = \oint_\gamma \frac{2z}{z^2+6z+1}\frac{dz}{iz} = \frac{2}{i}\oint_\gamma \frac{dz}{z^2+6z+1}$$

We have that the denominator has roots $z_\pm = -3\pm 2\sqrt{2}$. Since $z_+ z_- = 1$ and $|z_-| = 3+2\sqrt{2} > 1$, only $z_+ = -3+2\sqrt{2}$ lies inside γ, and it is a simple pole. Using Ex 9.2.3 (done above) and Lemm 9.16, with denominator z^2+6z+1 of derivative $2z+6$, we have,

$$\operatorname{Res}\left(\frac{1}{z^2+6z+1}; z_+\right) = \frac{1}{2z_+ + 6} = \frac{1}{2(-3+2\sqrt{2})+6} = \frac{1}{4\sqrt{2}}$$

Using Cor 9.13 (the Residue Theorem), we have,

$$J = \frac{2}{i} 2\pi i \frac{1}{4\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

Therefore, we have the integral evaluation as,

$$I = 2\pi - 3\cdot\frac{\pi}{\sqrt{2}} = 2\pi - \frac{3\sqrt{2}}{2}\pi = \frac{(4-3\sqrt{2})\pi}{2}$$

Ex 9.3.11

Evaluate
$$\int_0^\infty \frac{\cos(ax)}{(x^2+b^2)^2}\,dx \qquad \text{where } a>0 \text{ and } b>0.$$

Pf. We have that the integrand is even, therefore $\int_0^\infty = \frac{1}{2}\int_{-\infty}^\infty$. Let

$$f(z) = \frac{1}{(z^2+b^2)^2} = \frac{1}{(z-ib)^2(z+ib)^2}$$
$$g(z) = f(z)e^{iaz}$$

We have $f = p/q$ with $\deg p = 0$ and $\deg q = 4$, therefore $\deg p + 2 \leqslant \deg q$, and q has no real zeros. Since $a > 0$, we can use Prop 9.21 to get,

$$\int_{-\infty}^\infty \frac{e^{iax}}{(x^2+b^2)^2}\,dx = 2\pi i \sum_{\operatorname{Im} z>0} \operatorname{Res}(g;z) = 2\pi i \operatorname{Res}(g;ib)$$

where the only pole in the upper half-plane is the double pole at ib. Using Lemm 9.17, we have,

$$\operatorname{Res}(g;ib) = \lim_{z\to ib}\frac{d}{dz}\Big[(z-ib)^2 g(z)\Big] = \frac{d}{dz}\left[\frac{e^{iaz}}{(z+ib)^2}\right]_{z=ib}$$

We have,

$$\frac{d}{dz}\Big[e^{iaz}(z+ib)^{-2}\Big] = e^{iaz}(z+ib)^{-3}\big[\,ia(z+ib)-2\,\big]$$

At $z = ib$ we have $z+ib = 2ib$, $e^{iaz} = e^{-ab}$, $ia(2ib)-2 = -2(ab+1)$, and $(2ib)^{-3} = \dfrac{i}{8b^3}$. Therefore,

$$\operatorname{Res}(g;ib) = e^{-ab}\cdot\frac{i}{8b^3}\cdot(-2)(ab+1) = -\frac{i(ab+1)e^{-ab}}{4b^3}$$

and we get,

$$\int_{-\infty}^\infty \frac{e^{iax}}{(x^2+b^2)^2}\,dx = 2\pi i\cdot\left(-\frac{i(ab+1)e^{-ab}}{4b^3}\right) = \frac{\pi(1+ab)e^{-ab}}{2b^3}.$$

Using Rmk 9.23, we take the real parts and replace e^{iax} by $\cos(ax)$. The value above is already real, therefore,

$$\int_{-\infty}^\infty \frac{\cos(ax)}{(x^2+b^2)^2}\,dx = \frac{\pi(1+ab)e^{-ab}}{2b^3}$$

We can halve this to obtain the integral evaluation as,

$$\int_0^\infty \frac{\cos(ax)}{(x^2+b^2)^2}\,dx = \frac{\pi(1+ab)e^{-ab}}{4b^3}$$

Ex 9.3.12

Evaluate $\int_0^\infty \frac{1}{x^n+1}\,dx$ where $n \geqslant 2$. (Note that we cannot use the old reliable contour Γ_R from Proposition 9.21 since the integrand has a singularity at $x=-1$ when n is odd. Instead, I suggest you use the contour which is the positively oriented boundary of the circular sector $S = \{re^{i\theta} : 0 \leqslant \theta \leqslant \frac{2\pi}{n} \text{ and } 0 \leqslant r \leqslant \alpha\}$.)

Pf. Let $f(z) = \frac{1}{z^n+1}$ and let Γ be the positively oriented boundary of the sector S consisting of the segment $[0,\alpha]$ on the positive real axis, the arc C_α given by $z = \alpha e^{i\theta}$ for $0 \leqslant \theta \leqslant 2\pi/n$, and the segment back to the origin along the ray $\arg z = 2\pi/n$.

We have that the zeros of z^n+1 are $z = e^{i\pi(2k+1)/n}$. The only zero with argument in $(0, 2\pi/n)$ is $z_0 = e^{i\pi/n}$ (the next zero, $e^{i3\pi/n}$, has argument $3\pi/n > 2\pi/n$). Therefore z_0 is the unique pole inside the sector, and it is simple. Using Ex 9.2.3 (done above) and Lemm 9.16 (Simple Pole Lemma), with denominator z^n+1 of derivative nz^{n-1}, and using $z_0^n = -1$, we have,

$$\operatorname{Res}(f; z_0) = \frac{1}{nz_0^{n-1}} = \frac{z_0}{nz_0^n} = \frac{z_0}{-n} = -\frac{e^{i\pi/n}}{n}$$

Using Cor 9.13 (the Residue Theorem), we have,

$$\oint_\Gamma f(z)\,dz = 2\pi i \operatorname{Res}(f; z_0) = -\frac{2\pi i\, e^{i\pi/n}}{n}$$

We can now consider the three pieces for $\alpha \to \infty$:

- Outgoing segment: $\int_0^\alpha \frac{dx}{x^n+1} \to I$, which is the desired integral.
- Arc: Using Prop 3.26(a), since $n \geqslant 2$, we have,
$$\left|\int_{C_\alpha} f\right| \leqslant \frac{1}{\alpha^n - 1} \cdot \frac{2\pi\alpha}{n} \xrightarrow[\alpha\to\infty]{} 0$$
- Returning segmnet: We can paramatrize $z = re^{2\pi i/n}$ with r from α to 0. We get $z^n = r^n e^{2\pi i} = r^n$ and $dz = e^{2\pi i/n}\,dr$, therefore,
$$\int_\alpha^0 \frac{e^{2\pi i/n}}{r^n+1}\,dr = -e^{2\pi i/n}\int_0^\alpha \frac{dr}{r^n+1} \xrightarrow[\alpha\to\infty]{} -e^{2\pi i/n} I$$

We can add the three contributions and we see that as $\alpha \to \infty$,

$$I\left(1 - e^{2\pi i/n}\right) = -\frac{2\pi i\, e^{i\pi/n}}{n}$$

We can factor $1 - e^{2\pi i/n} = -e^{i\pi/n}\left(e^{i\pi/n} - e^{-i\pi/n}\right) = -e^{i\pi/n}\cdot 2i\sin(\pi/n)$. Therefore, we have,

$$I = \frac{-2\pi i\, e^{i\pi/n}}{-2i\, n\, e^{i\pi/n}\sin(\pi/n)} = \frac{\pi}{n\sin(\pi/n)}$$

which finally gives,

$$\int_0^\infty \frac{dx}{x^n+1} = \frac{\pi}{n\sin(\pi/n)}.$$

Ex 9.3.14

Use Proposition 9.33 with $f(z) = (z^2 - a^2)^{-1}$ to show that if $a \notin \mathbb{Z}$, then

$$\pi \cot(\pi a) = \frac{1}{a} + \sum_{n=1}^{\infty} \frac{2a}{a^2 - n^2}.$$

Pf. Let $f(z) = \dfrac{1}{z^2 - a^2} = \dfrac{p(z)}{q(z)}$ with $\deg p = 0$ and $\deg q = 2$, so that $\deg p + 2 \leqslant \deg q$. We have that its poles $\pm a$ are disjoint from $\mathbb{Z}$ because $a \notin \mathbb{Z}$. Therefore, we can use Prop 9.33 to $g(z) = \pi f(z) \cot(\pi z)$ to obtain,

$$\sum_{k=-\infty}^{\infty} \frac{1}{k^2 - a^2} = -\sum_{q(z)=0} \operatorname{Res}(g; z) = -\big(\operatorname{Res}(g; a) + \operatorname{Res}(g; -a)\big)$$

Each of these poles is simple. Since $a \notin \mathbb{Z}$, $\cot(\pi z)$ is holomorphic and finite at $\pm a$, so using Lemm 9.16 (Simple Pole Lemma) with denominator $z^2 - a^2$ of derivative $2z$, we have,

$$\operatorname{Res}(g; a) = \frac{\pi \cot(\pi a)}{2a}$$
$$\operatorname{Res}(g; -a) = \frac{\pi \cot(-\pi a)}{-2a} = \frac{\pi \cot(\pi a)}{2a}$$

using the fact that cot is odd. Therefore, we have,

$$\sum_{k=-\infty}^{\infty} \frac{1}{k^2 - a^2} = -\frac{\pi \cot(\pi a)}{a}$$

On LHS, we can separate $k = 0$ and pair k with $-k$ (for $k \geqslant 1$), using $k^2 = (-k)^2$, which gives,

$$\sum_{k=-\infty}^{\infty} \frac{1}{k^2 - a^2} = -\frac{1}{a^2} + \sum_{n=1}^{\infty} \frac{2}{n^2 - a^2}$$

Therefore, $-\dfrac{1}{a^2} + \sum_{n \geqslant 1} \dfrac{2}{n^2 - a^2} = -\dfrac{\pi \cot(\pi a)}{a}$. We can multiply by $-a$ and use $\dfrac{-2a}{n^2 - a^2} = \dfrac{2a}{a^2 - n^2}$ to get,

$$\pi \cot(\pi a) = \frac{1}{a} + \sum_{n=1}^{\infty} \frac{2a}{a^2 - n^2}$$

as required. □

Ex 11.1.3

Complete the proof of Lemma 11.8.

Lemma 11.8. Suppose that $|z| < \frac{1}{2}$. Then

$$\tfrac{1}{2}|z| \leqslant |\operatorname{Log}(1+z)| \leqslant \tfrac{3}{2}|z|.$$

Pf. From Maclaurin series $\operatorname{Log}(1+z) = \sum_{k\geqslant 1}(-1)^{k-1}z^k/k$, valid for $|z| < 1$. We can divide by z ($z \neq 0$) and compare coefficients with $\frac{1}{2}$. Comparing the bounds in equation (11.2): for $0 < |z| < 1$, we have,

$$\left|\frac{\operatorname{Log}(1+z)}{z} - 1\right| = \left|-\frac{z}{2} + \frac{z^2}{3} - \cdots\right| \leqslant \frac{1}{2}\left(|z| + |z|^2 + \cdots\right) = \frac{1}{2}\frac{|z|}{1-|z|}$$

We can complete the argument. Consider $0 < |z| < \frac{1}{2}$. The function $t \mapsto t/(1-t)$ is increasing on $[0,1)$, so

$$\frac{|z|}{1-|z|} < \frac{1/2}{1-1/2} = 1$$

when,

$$\left|\frac{\operatorname{Log}(1+z)}{z} - 1\right| \leqslant \frac{1}{2}$$

Using the (reverse) triangle inequality, we have,

$$1 - \frac{1}{2} \leqslant \left|\frac{\operatorname{Log}(1+z)}{z}\right| \leqslant 1 + \frac{1}{2}$$

i.e.

$$\frac{1}{2} \leqslant \left|\frac{\operatorname{Log}(1+z)}{z}\right| \leqslant \frac{3}{2}$$

We can multiply through by $|z| > 0$ to get,

$$\tfrac{1}{2}|z| \leqslant |\operatorname{Log}(1+z)| \leqslant \tfrac{3}{2}|z|$$

Using $z = 0$, the three quantities equal 0, so the inequalities hold there as well. This gives the assertion of Lemm 11.8.